

ThreatLens AI - Project Documentation

What This Is

ThreatLens AI is a full-stack security log analyzer and alert triage platform built for the Advanced Software Engineering course at University Metropolitan Tirana. The goal was to build something a real security team could use: a system that ingests logs, automatically flags suspicious activity, and helps analysts understand and respond to threats faster.

The platform supports three user roles with separate dashboards and access levels. Security Admins manage users and upload logs. SOC Analysts investigate alerts and run AI-powered triage. Viewers have read-only access to processed alerts and platform statistics.

Team and Work Division

The team had six members, each owning a specific part of the system:

Member	Responsibility
EK	Captain, AI research, prompt architecture, IoT classification coordination
AD	Backend structure, database schema, authentication, AI triage service
EI	Backend endpoints, analyst and viewer routes, external ingest API, IoT endpoints
GZ	Frontend scaffold, routing, design system, landing and login pages
FH	Role-based dashboards, IoT Classifier UI, Audit Log updates
SD	Ollama integration, prompt design, end-to-end validation, IoT model training

The backend was split between AD and EI: AD owned the project structure, database models, auth system, and the AI triage service layer. EI built the route handlers on top of that foundation and added the external log ingest API. On the frontend, GZ handled everything shared across the app (layout, routing, auth flow, design language) while FH built each individual dashboard.

Communication ran through a WhatsApp group with a daily check-in requirement. Task distribution for each week was coordinated by EK at the start of the week to avoid overlap between members working on connected parts of the system.

Tech Stack

Layer	Technology
Frontend	React, Vite, Tailwind CSS
Backend	Python, FastAPI
Database	PostgreSQL via SQLAlchemy
Anomaly Detection	scikit-learn Isolation Forest
IoT Classification	LightGBM
AI Triage	Ollama (local LLM, llama3.1:8b)

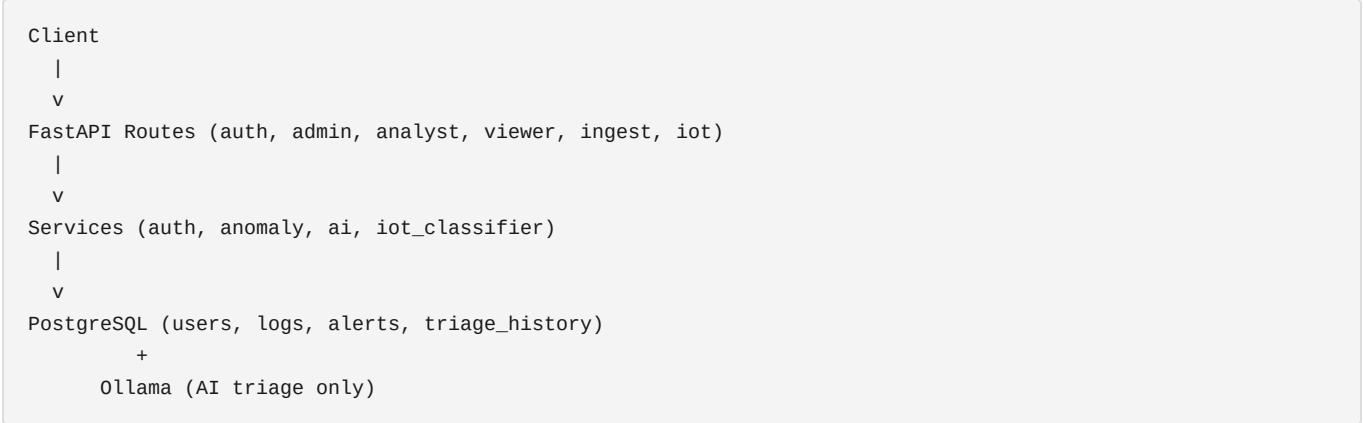
The decision to use a local LLM through Ollama rather than an external API was deliberate. It keeps all data on the network, avoids API costs, and removes the dependency on an internet connection during a demo or classroom setting. The tradeoff is that the model needs to be pulled and running locally.

Isolation Forest was chosen for log anomaly detection because it works without labeled training data. Every deployment has different log patterns, so a model that learns what normal looks like for a given log file and flags deviations from that is more practical than one trained on a fixed labeled dataset.

LightGBM was used for the IoT classifier because the feature space is wide (100 columns after selection) and the dataset is tabular. It handles that well and inference is fast enough to be done inline during a request.

Architecture

The backend is a FastAPI application organized into routes, services, and models. Routes handle HTTP concerns and call into services for business logic. Services have no knowledge of HTTP. The database session is injected via a dependency at the route level.



The frontend is a React SPA with client-side routing. An Axios instance attaches the JWT token to every outgoing request. Protected routes check the role from context before rendering and redirect unauthenticated users to login.

Database

Four tables:

users - registered accounts with hashed passwords and a role (admin, analyst, viewer).

logs - uploaded log files stored as raw text with a filename, source label, and a reference to the uploader.

alerts - flagged entries from log analysis or IoT submissions. Each alert has a severity, a status (open, triaged, resolved), the flagged message, and an optional malware_family field. The log_id foreign key is nullable because IoT alerts are not linked to a log file.

trriage_history - records of analyst triage actions including the AI-generated summary and remediation steps.

Schema migrations for new columns run automatically on startup using idempotent `ALTER TABLE IF NOT EXISTS` statements, so there is no manual migration step needed when pulling updates.

Core Features

Log Ingestion and Anomaly Detection

Logs can be uploaded through the Admin dashboard or pushed directly via the ingest API. Either way, the backend stores the raw content and immediately runs anomaly detection.

Each line of the log is turned into a four-dimensional feature vector covering line length, error keyword count, IP address count, and port number presence. Isolation Forest is fit on all lines and flags outliers. The anomaly score is adjusted by the error keyword weight and mapped to a severity level. An Alert record is created for each flagged line.

AI Alert Triage

When an analyst triggers triage, the backend builds a context window of five lines above and below the flagged line in the original log. That context, the alert message, and the severity level are sent to Ollama as a prompt requesting a structured JSON response with a plain-language summary and concrete remediation steps. The result is stored and the alert status is updated to triaged.

IoT Malware Classification

The IoT classifier uses a LightGBM model trained on the CIC-YNU-IoTMal dataset across four architectures (arm, mips, mipsel, x86) using behavioral features from network captures, system activity reports, and system call traces. It classifies submissions into five families: Benign, Mirai, DarkNexus, Gafgyt, and Generic.

Severity is derived from the predicted family. Mirai and DarkNexus map to critical, Gafgyt and Generic to high, Benign to low. If the confidence score is below 50 percent the severity is capped at medium. If the model artifacts are not present the endpoint still accepts submissions but marks the result as unclassified.

The model artifacts are not included in the repository. To generate them, run all cells in `iot_malware_detection.ipynb` with the CIC-YNU-IoTMal dataset placed at `data/work/samples/`.

UI

Landing Page



Login

THREATLENS AI

OPERATOR AUTHENTICATION

secret_login.sh

IDENTIFIER

operator@domain

PASSPHRASE

SHOW

AUTHENTICATE

RETURN TO HOME

Admin Dashboard

THREATLENS

SECURITY ADMIN

Overview

Users

Upload Logs

Audit Log

IoT Classifier

admin@threatlens.local

SIGN OUT

Platform Overview

Alert distribution across all ingested log sources

TOTAL ALERTS	OPEN	TRIAGED	RESOLVED
10	5	5	0

● THREATLENS

SECURITY ADMIN

Overview

Users

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Audit Log

IoT Classifier

admin@threatens.local

SIGN OUT

User Management

Registered operators and their assigned roles

EMAIL

PASSWORD

ROLE

CREATE USER

EMAIL	ROLE	REGISTERED	
test@test.com	VIEWER	5/9/2026	REMOVE
admin@threatens.local	ADMIN	5/9/2026	REMOVE
analyst@threatens.local	ANALYST	5/9/2026	REMOVE
viewer@threatens.local	VIEWER	5/9/2026	REMOVE

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SECURITY ADMIN

Overview

Users

Upload Logs

Audit Log

IoT Classifier

admin@threatens.local

SIGN OUT

Ingest Log File

Upload a plaintext log file for anomaly analysis

SOURCE

nginx / syslog / firewall

FILE (.txt / .log)

Choose File No file chosen

INGEST LOG

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SECURITY ADMIN

Overview

Users

Upload Logs

Audit Log

IoT Classifier

admin@threatlens.local

SIGN OUT

Audit Log

All platform alerts ordered by creation time

ID	SEVERITY	FAMILY	STATUS	MESSAGE	TIME STAMP
9	CRITICAL		triaged	2026-05-11 09:13:30 CRITICAL firewall: SPOOF DETECTED inbound TCP from ...	5/11/2026, 10:39:06 AM
8	CRITICAL		open	2026-05-11 09:07:18 INFO firewall: ALLOWED outbound UDP from 10.0.0.5:123 ...	5/11/2026, 10:39:06 AM
7	CRITICAL		open	2026-05-11 09:05:00 INFO firewall: ALLOWED inbound ICMP from 192.168.1.1 L...	5/11/2026, 10:39:06 AM
10	CRITICAL		triaged	2026-05-11 09:15:00 INFO firewall: ALLOWED inbound ICMP from 192.168.1.1 L...	5/11/2026, 10:39:06 AM
6	CRITICAL		open	2026-05-09 08:09:45 ERROR nginx: upstream timed out connecting to 127.0.0.1:8000	5/9/2026, 10:41:55 AM
4	CRITICAL		open	2026-05-09 08:04:25 CRITICAL firewall: DENIED inbound connection from 45.33...	5/9/2026, 10:41:55 AM
5	CRITICAL		open	2026-05-09 08:04:26 CRITICAL firewall: DENIED inbound connection from 45.33...	5/9/2026, 10:41:55 AM
1	CRITICAL		triaged	2026-05-09 08:04:25 CRITICAL firewall: DENIED inbound connection from 45.33...	5/9/2026, 10:28:26 AM
3	CRITICAL		triaged	2026-05-09 08:09:45 ERROR nginx: upstream timed out connecting to 127.0.0.1:8000	5/9/2026, 10:28:26 AM
2	CRITICAL		triaged	2026-05-09 08:04:26 CRITICAL firewall: DENIED inbound connection from 45.33...	5/9/2026, 10:28:26 AM

● THREATLENS

SECURITY ADMIN

Overview

Users

Upload Logs

Audit Log

IoT Classifier

admin@threatlens.local

SIGN OUT

IoT Malware Classifier

Submit behavioural feature vectors for LightGBM family classification

FEATURE PAYLOAD (JSON)

```
{
  "hash": "abc123",
  "arch": "arm",
  "features": [
    "strace_Call_connect": 12500,
    "strace_Call_send": 8200,
    "pcap_Rate_mean": 950.0
  ]
}
```

SUBMIT SAMPLE

Analyst Dashboard

● THREATLENS

SOC ANALYST

Active Alerts

Log Explorer

Triage Console

Triage History

analyst@threatlens.local

SIGN OUT

Active Alerts

Click any alert to open it in the Triage Console

ID	SEVERITY	FAMILY	STATUS	MESSAGE	TIME STAMP
9	CRITICAL		triaged	2026-05-11 09:13:30 CRITICAL firewall: SPOOF DETECTED inbound TCP from...	5/11/2026, 10:39:06 AM
8	CRITICAL		open	2026-05-11 09:07:18 INFO firewall: ALLOWED outbound UDP from 10.0.0.5:12...	5/11/2026, 10:39:06 AM
7	CRITICAL		open	2026-05-11 09:05:00 INFO firewall: ALLOWED inbound ICMP from 192.168.1.1 L...	5/11/2026, 10:39:06 AM
10	CRITICAL		triaged	2026-05-11 09:15:00 INFO firewall: ALLOWED inbound ICMP from 192.168.1.1 L...	5/11/2026, 10:39:06 AM
6	CRITICAL		open	2026-05-09 08:09:45 ERROR nginx: upstream timed out connecting to 127.0....	5/9/2026, 10:41:55 AM
4	CRITICAL		open	2026-05-09 08:04:25 CRITICAL firewall: DENIED inbound connection from 4...	5/9/2026, 10:41:55 AM
5	CRITICAL		open	2026-05-09 08:04:26 CRITICAL firewall: DENIED inbound connection from 4...	5/9/2026, 10:41:55 AM
1	CRITICAL		triaged	2026-05-09 08:04:25 CRITICAL firewall: DENIED inbound connection from 4...	5/9/2026, 10:26:26 AM
3	CRITICAL		triaged	2026-05-09 08:09:45 ERROR nginx: upstream timed out connecting to 127.0....	5/9/2026, 10:26:26 AM
2	CRITICAL		triaged	2026-05-09 08:04:26 CRITICAL firewall: DENIED inbound connection from 4...	5/9/2026, 10:26:26 AM

● THREATLENS

SOC ANALYST

Active Alerts

Log Explorer

Triage Console

Triage History

analyst@threatlens.local

SIGN OUT

Triage Console

Run AI-assisted triage on an alert via local LLM

Alert ID

RUN TRIAGE

● THREATLENS
SOC ANALYST

Active Alerts

Log Explorer

Triage Console

Triage History

Triage History

All completed AI triage sessions

ALERT #2511/2026, 10:50:36 AM

SUMMARY

An unknown entity is attempting to connect to port 443 on our server, which may be a reconnaissance attempt for further exploitation. The firewall has denied the connection, but we should investigate why the entity was able to generate SYN flood warnings before being blocked.

REMEDIAION

Review firewall logs and network traffic to identify the source of the attempted connections. Check for any open ports or services that may have been exploited. Consider implementing rate limiting on incoming connections to prevent future SYN floods. Also, consider generating an alert when a connection is made from a new IP address.

ALERT #9511/2026, 10:56:46 AM

SUMMARY

A critical spoofing attack has been detected, where the source IP address is identical to the destination IP address. This indicates potential unauthorized access or data manipulation.

REMEDIAION

Immediately investigate and identify the malicious process on 10.0.0.5:9999, then terminate it and review firewall rules for vulnerabilities that allowed this attack.

ALERT #9511/2026, 10:52:25 AM

SUMMARY

A spoofing attack has been detected where the source IP and destination IP are the same (10.0.0.5:9999 to 10.0.0.5:443), indicating potential malicious activity.

REMEDIAION

Check the firewall rules for any anomalies or misconfigurations, and review system logs for potential evidence of unauthorized access or data exfiltration.

ALERT #10511/2026, 10:49:20 AM

SUMMARY

A potential security threat is detected due to an ICMP packet from a trusted host allowing for potential spoofing and information disclosure, which may be a precursor to a larger attack. Further investigation is required to understand the context of this event.

REMEDIAION

Block all inbound ICMP traffic at the firewall level, review logs for any suspicious activity related to 192.168.1.1, and perform a thorough vulnerability assessment on the affected system.

ALERT #9511/2026, 10:40:25 AM

SUMMARY

A spoofing attack has been detected where an IP is attempting to connect to itself, indicating potential malicious activity.

REMEDIAION

Immediately block all incoming traffic on the destination port (443) and investigate the source of the spoofed packet for further action.

ALERT #2519/2026, 10:45:27 AM

SUMMARY

The firewall has denied an inbound connection from IP 45.33.32.156 on port 4444 to port 443, indicating potential scanning or exploitation attempts.

REMEDIAION

Check the firewall logs for more information about the denied connection and investigate why this IP is attempting to connect. Consider blocking 45.33.32.156 at the firewall level if it's not a legitimate traffic source.

Viewer Dashboard

● THREATLENS
VIEWER

Dashboard

Alerts

Security Dashboard

Platform-wide alert status summary

TOTAL	OPEN	TRIAGED	RESOLVED
10	5	5	0

viewer@threatlens.local

SIGN OUT

● THREATLENS

VIEWER

Dashboard

Alerts

viewer@threatiens.local

SIGN OUT

Alerts

Triaged and resolved alerts visible to this role

ID	SEVERITY	STATUS	MESSAGE	TIME STAMP
9	CRITICAL	triaged	2026-05-11 09:13:30 CRITICAL firewall: SPOOF DETECTED inbound TCP from 10.0.0.5:9999 ...	5/11/2026, 10:39:06 AM
10	CRITICAL	triaged	2026-05-11 09:15:00 INFO firewall: ALLOWED inbound ICMP from 192.168.1.1 to 10.0.0.5	5/11/2026, 10:39:06 AM
1	CRITICAL	triaged	2026-05-09 08:04:25 CRITICAL firewall: DENIED inbound connection from 45.33.32.156:44...	5/9/2026, 10:28:26 AM
2	CRITICAL	triaged	2026-05-09 08:04:26 CRITICAL firewall: DENIED inbound connection from 45.33.32.156:44...	5/9/2026, 10:28:26 AM
3	CRITICAL	triaged	2026-05-09 08:09:45 ERROR nginx: upstream timed out connecting to 127.0.0.1:8000	5/9/2026, 10:28:26 AM

Setup

Prerequisites

- Python 3.11 or higher
- Node.js 18 or higher
- PostgreSQL
- Ollama with llama3.1:8b pulled

```
ollama pull llama3.1:8b
```

Backend

```
cd backend
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
cp .env.example .env
```

Fill in `.env` with your database URL, a secret key, and the Ollama URL. Then start the server:

```
uvicorn app.main:app --port 8000 --reload
```

Frontend

```
cd frontend
npm install
npm run dev
```

First User

```
curl -X POST http://localhost:8000/auth/register \  
-H "Content-Type: application/json" \  
-d '{"email": "admin@yourorg.com", "password": "yourpassword", "role": "admin"}'
```

After that, additional users can be created from the Admin dashboard.

Branching

Feature branches follow the naming convention `feature/name-or-feature`. Do not commit directly to main. Open a pull request and get at least one review before merging.